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FACULTY OF ENGINEERING
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Test Plan, Test Design Specifications and Test Cases

Version 1

CENG 408

Innovative System Design and Development II

IoT Integrated Accessible Smart Home System

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1. INTRODUCTION

1.1 Version Control

Version No	Description of Changes	Date
1.0	Initial completed test plan prepared from the project wiki, implementation notes, system requirements, and validation page.	June 04, 2026

1.2 Overview

This document defines the test plan, test design specifications, and detailed test cases for the IoT Integrated Accessible Smart Home System. The system is an accessible smart home control platform that converts computer-vision events such as eye gaze, blink gestures, hand gestures, and face-authentication outcomes into backend actions and physical smart device commands.

The tested architecture contains a Raspberry Pi 5 vision-processing module, a FastAPI backend, a React Native Expo mobile application, and TP-Link/Tapo smart device integration over a local network. The core end-to-end behavior to validate is: User Gesture -> Pi Vision Detection -> Backend Event Processing -> Action Mapping -> Device Command -> Physical Device Response.

1.3 Scope

This test plan covers functional, integration, device-control, accessibility, privacy, and reliability testing for the implemented prototype. It focuses on the current deliverable: local-network operation, eye/blink event generation, backend mapping and execution, mobile monitoring/configuration, mock and real smart bulb/plug behavior, and final end-to-end validation.

- In scope: backend API endpoints, event validation, mapping updates, action execution, UI-state navigation, device-state updates, mock mode, real TP-Link/Tapo light control, Raspberry Pi camera input, MediaPipe-based gesture detection, mobile API communication, accessible mode behavior, and end-to-end smart home response.
- Out of scope for this version: commercial deployment certification, large-scale multi-user testing, cloud-based remote access, Matter/Zigbee/Z-Wave device ecosystems, database persistence, and production security penetration testing.

1.4 Terminology

Acronym	Definition
APR	Accessibility, Privacy and Reliability
API	Application Programming Interface
BE	Backend API
EAR	Eye Aspect Ratio
EI	End-to-End Integration
ESP32	WiFi-enabled microcontroller used for IoT control expansion
FastAPI	Python web framework used for the backend API
IoT	Internet of Things
Kasa/Tapo	TP-Link smart home device ecosystem
LAN	Local Area Network
MA	Mobile Application
MediaPipe	Computer vision framework used for Face Mesh, iris/eye landmarks and hand detection
SD	Smart Device Control
TTS	Text-to-Speech
VP	Vision Processing and Gesture Detection

2. FEATURES TO BE TESTED

2.1 Backend API (BE)

FastAPI service startup, health/status reporting, supported event/action discovery, mapping retrieval/update, action execution, validation errors, and backward-compatible event endpoints.

2.2 Vision Processing and Gesture Detection (VP)

IP camera stream opening, frame preprocessing, MediaPipe landmark extraction, gaze classification, blink duration classification, hand gesture events, face-authentication event flow, and backend-compatible event payload generation.

2.3 Mobile Application (MA)

React Native Expo app connection to the backend, dashboard/status display, accessible UI mode, mapping configuration, manual action execution, test event sending, error handling, and Turkish spoken feedback.

2.4 Smart Device Control (SD)

Mock device mode, real TP-Link/Tapo smart bulb and plug configuration, LIGHT_ON/LIGHT_OFF, PLUG_ON/PLUG_OFF behavior, hardware response reporting, and no-crash behavior when hardware is absent.

2.5 End-to-End Integration (EI)

Complete local-network flow from user gesture to Raspberry Pi event, backend mapping, action execution, mobile state update, and physical device response.

2.6 Accessibility, Privacy and Reliability (APR)

Hands-free usability, non-verbal interaction, local processing/privacy constraints, graceful disconnection handling, false-trigger reduction, and real-time responsiveness on Raspberry Pi 5.

3. FEATURES NOT TO BE TESTED

Feature Not Tested	Reason
Large-scale multi-user control	The current pipeline assumes one active face/hand/user at a time; concurrent multi-user control is future work.
Remote/cloud access	The prototype is designed for local LAN operation without cloud dependency; remote access is future work.
Matter, Zigbee, Z-Wave ecosystems	The current implementation is focused on TP-Link/Tapo devices and mock prototype actions.
Persistent database layer	Backend state is in memory in the current implementation; database persistence is future work.
Formal WCAG certification	Accessible design is tested functionally, but formal WCAG audit/ certification is outside this course prototype scope.
Long-term reliability/aging tests	Continuous multi-day hardware stress testing is outside the schedule of this test plan.
Electrical certification	Prototype device behavior is tested, but professional electrical compliance certification is outside the scope.

4. ITEM PASS/FAIL CRITERIA

A test case passes when the actual result matches the expected result, no unhandled exception or crash occurs, and the system state remains valid after the operation. A test case fails when any required response field, UI update, device-state change, validation error, or physical hardware response does not match the expected behavior.

- High priority tests must pass for the release candidate to be accepted.
- Medium priority tests should pass unless a documented workaround exists.
- Low priority tests may be deferred only if they do not block the main demonstration flow.
- For hardware tests, both backend state and observed physical device response must be checked when real hardware is configured.
- For vision tests, false positives and repeated accidental triggers must remain within an acceptable manual observation level during demonstration scenarios.

4.1 Exit Criteria

- 100% of High priority test cases are executed.
- At least 95% of all listed test cases pass.
- All High and Medium priority backend, vision, mobile, smart-device, and end-to-end test cases pass.

- No known crash remains in the final demo path: gesture detection -> backend mapping -> action execution -> device response.
- Mock mode and real smart bulb mode are both verified.
- Open issues, if any, are documented with severity and workaround.

5. REFERENCES

[1] GitHub Wiki, 01. Home, IoT Integrated Accessible Smart Home System.

[2] GitHub Wiki, 02. Introduction & Objectives.

[3] GitHub Wiki, 04. System Requirements.

[4] GitHub Wiki, 05. System Architecture.

[5] GitHub Wiki, 06. Implementation.

[6] GitHub Wiki, 07. Testing & Validation.

[7] GitHub Wiki, 09. Conclusion & Future Work.

[8] Sample_Test_Plan.docx template uploaded by the project team.

6. TEST DESIGN SPECIFICATIONS

6.1 Backend API (BE)

6.1.1 Subfeatures to be tested

Backend Startup - Server runs without runtime errors and exposes the documented endpoints.

Event and Action Processing - Backend validates events, maps them to actions, updates state, and returns consistent JSON responses.

Mapping Configuration - Runtime mapping updates are accepted only for supported events and actions.

6.1.2 Test Cases

TC ID	Requirements	Priority	Scenario Description
BE.ST.01	FR-API-01	High	Start FastAPI server and verify root endpoint returns API running message.
BE.HL.01	FR-API-02	High	Call /health and verify success, controller mode, and UI snapshot.
BE.STA.01	FR-API-03	High	Call /status and verify system state, UI state, device status, controller mode, and mappings.
BE.EVT.01	FR-API-04	High	Call /vision-event-types and verify four core eye events are returned.
BE.ACT.01	FR-API-05	High	Call /actions and verify navigation and device actions are returned.
BE.MAP.01	FR-API-06	High	Retrieve default mappings for look_left, look_right, short_blink, and long_blink.
BE.MAP.02	FR-API-07	High	Update a valid event-to-action mapping and verify the change is stored.
BE.MAP.03	FR-API-08	High	Submit invalid event/action mapping and verify 400 Bad
BE.EXE.01	FR-API-09	High	Manually execute NAV_RIGHT and verify selected UI item changes.
BE.EVT.02	FR-API-10	High	POST valid look_right event to /vision-events and verify mapped action executes.
BE.EVT.03	FR-API-11	High	POST unsupported vision event and verify it is rejected without corrupting state.
BE.BC.01	FR-API-12	Medium	POST to backward-compatible /event endpoint and verify same behavior as /vision-events.

6.2 Vision Processing and Gesture Detection (VP)

6.2.1 Subfeatures to be tested

Camera Capture - Configurable camera stream is opened and processed locally on Raspberry Pi.

Eye Gesture Detection - MediaPipe landmarks are converted into gaze and blink events.

Advanced Vision Events - Hand gesture recognition and face-authentication behavior are validated for the final multi-modal system.

6.2.2 Test Cases

TC ID	Requirements	Priority	Scenario Description
VP.CAM.01	FR-VP-01	High	Open configured IP camera URL and read frames continuously.
VP.CAM.02	FR-VP-02	High	Handle unreachable camera stream safely without crash.
VP.FR.01	FR-VP-03	Medium	Resize frame and convert BGR to RGB before MediaPipe processing.
VP.LM.01	FR-VP-04	High	Detect one face and obtain eye/iris landmarks in normal lighting.
VP.GZ.01	FR-VP-05	High	Generate look_left after stable left gaze duration.
VP.GZ.02	FR-VP-06	High	Generate look_right after stable right gaze duration.
VP.GZ.03	NFR-VP-01	Medium	Avoid triggering gaze events for short accidental eye movements.
VP.BL.01	FR-VP-07	High	Generate short_blink within configured short-blink duration range.
VP.BL.02	FR-VP-08	High	Generate long_blink within configured long-blink duration range.
VP.PL.01	FR-VP-09	High	Publish backend-compatible event payload with source, name, confidence, timestamp, and metadata.
VP.HD.01	FR-VP-10	Medium	Recognize supported hand gestures used by the final system.
VP.FA.01	FR-VP-11	Medium	Verify face-authentication allows only recognized authorized user in the control flow.

6.3 Mobile Application (MA)

6.3.1 Subfeatures to be tested

Connection and Monitoring - Mobile application communicates with backend API and displays state.

Configuration and Testing - Users can update mappings, send test events, and execute actions manually.

Accessibility Interface - Accessible mode and speech feedback are validated.

6.3.2 Test Cases

TC ID	Requirements	Priority	Scenario Description
MA.CON.01	FR-MA-01	High	Mobile app connects to backend using EXPO_PUBLIC_API_URL.
MA.STA.01	FR-MA-02	High	Fetch /status and display system/ device/UI state correctly.
MA.EVT.01	FR-MA-03	High	Fetch supported vision event types and display them for test/ configuration.
MA.ACT.01	FR-MA-04	High	Fetch supported backend actions and display available action options.
MA.MAP.01	FR-MA-05	High	Retrieve current mappings in the configuration screen.
MA.MAP.02	FR-MA-06	High	Update mappings from the app and verify backend reflects the new mapping.
MA.TEST.01	FR-MA-07	High	Send a test vision event from the app and verify backend action execution.
MA.MAN.01	FR-MA-08	High	Manually execute LIGHT_ON/ LIGHT_OFF actions from the app.
MA.ACC.01	NFR-MA-01	Medium	Verify accessible mode with enlarged controls, high contrast layout, and understandable labels.
MA.TTS.01	NFR-MA-02	Medium	Verify Turkish TTS feedback is spoken after device actions.
MA.ERR.01	NFR-MA-03	Medium	Verify backend connection failure is shown to the user without app crash.

6.4 Smart Device Control (SD)

6.4.1 Subfeatures to be tested

Mock Mode - Backend remains usable without physical hardware.

Real Device Mode - TP-Link/Tapo smart bulb responds to commands through python-kasa.

Prototype Device State - Plug commands update backend state even where real binding is not available.

6.4.2 Test Cases

TC ID	Requirements	Priority	Scenario Description
SD.MOCK.01	FR-SD-01	High	Run backend without SMART_BULB_HOST and verify mock mode works.
SD.CFG.01	FR-SD-02	High	Configure SMART_BULB_HOST and credentials and verify controller mode is python-kasa.
SD.LGT.01	FR-SD-03	High	Execute LIGHT_ON and verify backend state and physical bulb on response.
SD.LGT.02	FR-SD-04	High	Execute LIGHT_OFF and verify backend state and physical bulb off response.
SD.PLG.01	FR-SD-05	Medium	Execute PLUG_ON and verify backend plug state updates.
SD.PLG.02	FR-SD-06	Medium	Execute PLUG_OFF and verify backend plug state updates.
SD.NOB.01	NFR-SD-01	Medium	Execute device actions when hardware is absent and verify no crash occurs.

6.5 End-to-End Integration (EI)

6.5.1 Subfeatures to be tested

Gesture to Intent - Vision events travel from Raspberry Pi to backend and become actions.

Intent to Device - Mapped actions update UI/device state and produce physical smart device feedback.

Runtime Reconfiguration - Mobile mapping changes affect the next end-to-end event without code changes.

6.5.2 Test Cases

TC ID	Requirements	Priority	Scenario Description
EI.EYE.01	FR-EI-01	High	Eye look_right event navigates UI and updates mobile status.
EI.EYE.02	FR-EI-02	High	Short blink selects Light menu and executes configured action.
EI.EYE.03	FR-EI-03	High	Long blink returns to main menu or cancels current selection.
EI.DEV.01	FR-EI-04	High	Gesture mapped to LIGHT_ON turns real smart bulb on.
EI.DEV.02	FR-EI-05	High	Gesture mapped to LIGHT_OFF turns real smart bulb off.
EI.MAP.01	FR-EI-06	High	Update mapping in mobile app and verify next Raspberry Pi event uses new mapping.
EI.HND.01	FR-EI-07	Medium	Hand gesture event triggers mapped navigation/action in final system.
EI.AUTH.01	FR-EI-08	Medium	Unauthorized face does not gain device-control access.

6.6 Accessibility, Privacy and Reliability (APR)

6.6.1 Subfeatures to be tested

Real-Time Operation - System runs responsively on Raspberry Pi 5 and LAN.

Privacy and Local Processing - Camera data remains local and the architecture avoids cloud dependency.

Fault Tolerance - Camera, backend, and network failures are handled without crashes.

6.6.2 Test Cases

TC ID	Requirements	Priority	Scenario Description
APR.RT.01	NFR-APR-01	High	Verify gesture-to-action response feels real-time on Raspberry Pi 5 during demo.
APR.PR.01	NFR-APR-02	High	Verify camera frames are processed locally and not uploaded to cloud services.
APR.DIS.01	NFR-APR-03	Medium	Disconnect camera stream and verify graceful error handling.
APR.NET.01	NFR-APR-04	Medium	Disconnect backend/network and verify mobile app shows connection error.
APR.FP.01	NFR-APR-05	High	Verify stable-gaze duration and blink thresholds reduce false triggers.
APR.ACC.01	NFR-APR-06	Medium	Verify hands-free use path is achievable without physical buttons.
APR.LOG.01	NFR-APR-07	Low	Verify FPS, gaze state, and event logs are available for debugging.

7. DETAILED TEST CASES

7.1 BE.ST.01

Field	Description
TC_ID	BE.ST.01
Purpose	Start FastAPI server and verify root endpoint returns API running message.
Requirements	FR-API-01
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Install backend dependencies and run Uvicorn/FastAPI server.
Procedure	1. Start backend. 2. Open root endpoint /. 3. Observe response body and server logs.
Expected Result / Verification	Server starts successfully and root endpoint returns an API running message.
Cleanup	Return system to default state if needed.

7.2 BE.HL.01

Field	Description
TC_ID	BE.HL.01
Purpose	Call /health and verify success, controller mode, and UI snapshot.
Requirements	FR-API-02
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Call /health and verify success, controller mode, and UI snapshot. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.3 BE.STA.01

Field	Description
TC_ID	BE.STA.01
Purpose	Call /status and verify system state, UI state, device status, controller mode, and mappings.
Requirements	FR-API-03
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Call /status and verify system state, UI state, device status, controller mode, and mappings. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.4 BE.EVT.01

Field	Description
TC_ID	BE.EVT.01
Purpose	Call /vision-event-types and verify four core eye events are returned.
Requirements	FR-API-04
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Call /vision-event-types and verify four core eye events are returned. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.5 BE.ACT.01

Field	Description
TC_ID	BE.ACT.01
Purpose	Call /actions and verify navigation and device actions are returned.
Requirements	FR-API-05
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Call /actions and verify navigation and device actions are returned. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.6 BE.MAP.01

Field	Description
TC_ID	BE.MAP.01
Purpose	Retrieve default mappings for look_left, look_right, short_blink, and long_blink.
Requirements	FR-API-06
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Retrieve default mappings for look_left, look_right, short_blink, and long_blink. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.7 BE.MAP.02

Field	Description
TC_ID	BE.MAP.02
Purpose	Update a valid event-to-action mapping and verify the change is stored.
Requirements	FR-API-07
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Send PUT /mappings with {"short_blink":"LIGHT_ON"}. 2. Call GET /mappings. 3. Send POST /vision-events using short_blink.
Expected Result / Verification	Mapping is updated and later short_blink event executes LIGHT_ON.
Cleanup	Return system to default state if needed.

7.8 BE.MAP.03

Field	Description
TC_ID	BE.MAP.03
Purpose	Submit invalid event/action mapping and verify 400 Bad Request.
Requirements	FR-API-08
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Submit invalid event/action mapping and verify 400 Bad Request. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.9 BE.EXE.01

Field	Description
TC_ID	BE.EXE.01
Purpose	Manually execute NAV_RIGHT and verify selected UI item changes.
Requirements	FR-API-09
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Manually execute NAV_RIGHT and verify selected UI item changes. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.10 BE.EVT.02

Field	Description
TC_ID	BE.EVT.02
Purpose	POST valid look_right event to /vision-events and verify mapped action executes.
Requirements	FR-API-10
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: POST valid look_right event to /vision-events and verify mapped action executes. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.11 BE.EVT.03

Field	Description
TC_ID	BE.EVT.03
Purpose	POST unsupported vision event and verify it is rejected without corrupting state.
Requirements	FR-API-11
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: POST unsupported vision event and verify it is rejected without corrupting state. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.12 BE.BC.01

Field	Description
TC_ID	BE.BC.01
Purpose	POST to backward-compatible /event endpoint and verify same behavior as /vision-events.
Requirements	FR-API-12
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: POST to backward-compatible /event endpoint and verify same behavior as /vision-events. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.13 VP.CAM.01

Field	Description
TC_ID	VP.CAM.01
Purpose	Open configured IP camera URL and read frames continuously.
Requirements	FR-VP-01
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Open configured IP camera URL and read frames continuously. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.14 VP.CAM.02

Field	Description
TC_ID	VP.CAM.02
Purpose	Handle unreachable camera stream safely without crash.
Requirements	FR-VP-02
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Handle unreachable camera stream safely without crash. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.15 VP.FR.01

Field	Description
TC_ID	VP.FR.01
Purpose	Resize frame and convert BGR to RGB before MediaPipe processing.
Requirements	FR-VP-03
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Resize frame and convert BGR to RGB before MediaPipe processing. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.16 VP.LM.01

Field	Description
TC_ID	VP.LM.01
Purpose	Detect one face and obtain eye/iris landmarks in normal lighting.
Requirements	FR-VP-04
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Detect one face and obtain eye/iris landmarks in normal lighting. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.17 VP.GZ.01

Field	Description
TC_ID	VP.GZ.01
Purpose	Generate look_left after stable left gaze duration.
Requirements	FR-VP-05
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Camera positioned toward one user under normal lighting; vision module running.
Procedure	1. Look left and hold gaze for at least configured stable duration. 2. Watch event log and payload publisher.
Expected Result / Verification	A single look_left event is generated and payload uses backend-supported event name.
Cleanup	Return system to default state if needed.

7.18 VP.GZ.02

Field	Description
TC_ID	VP.GZ.02
Purpose	Generate look_right after stable right gaze duration.
Requirements	FR-VP-06
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Generate look_right after stable right gaze duration. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.19 VP.GZ.03

Field	Description
TC_ID	VP.GZ.03
Purpose	Avoid triggering gaze events for short accidental eye movements.
Requirements	NFR-VP-01
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Avoid triggering gaze events for short accidental eye movements. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.20 VP.BL.01

Field	Description
TC_ID	VP.BL.01
Purpose	Generate short_blink within configured short-blink duration range.
Requirements	FR-VP-07
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Perform a deliberate short blink within configured short-blink timing. 2. Observe event log and payload.
Expected Result / Verification	short_blink event is generated; involuntary normal blinks do not repeatedly trigger commands.
Cleanup	Return system to default state if needed.

7.21 VP.BL.02

Field	Description
TC_ID	VP.BL.02
Purpose	Generate long_blink within configured long-blink duration range.
Requirements	FR-VP-08
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Generate long_blink within configured long-blink duration range. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.22 VP.PL.01

Field	Description
TC_ID	VP.PL.01
Purpose	Publish backend-compatible event payload with source, name, confidence, timestamp, and metadata.
Requirements	FR-VP-09
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Publish backend-compatible event payload with source, name, confidence, timestamp, and metadata. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.23 VP.HD.01

Field	Description
TC_ID	VP.HD.01
Purpose	Recognize supported hand gestures used by the final system.
Requirements	FR-VP-10
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Recognize supported hand gestures used by the final system. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.24 VP.FA.01

Field	Description
TC_ID	VP.FA.01
Purpose	Verify face-authentication allows only recognized authorized user in the control flow.
Requirements	FR-VP-11
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify face-authentication allows only recognized authorized user in the control flow. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.25 MA.CON.01

Field	Description
TC_ID	MA.CON.01
Purpose	Mobile app connects to backend using EXPO_PUBLIC_API_URL.
Requirements	FR-MA-01
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Backend running; mobile app configured with EXPO_PUBLIC_API_URL.
Procedure	1. Open mobile app. 2. Trigger status refresh. 3. Observe connection state.
Expected Result / Verification	Application receives backend status within timeout and does not show connection error.
Cleanup	Return system to default state if needed.

7.26 MA.STA.01

Field	Description
TC_ID	MA.STA.01
Purpose	Fetch /status and display system/device/UI state correctly.
Requirements	FR-MA-02
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Fetch /status and display system/device/UI state correctly. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.27 MA.EVT.01

Field	Description
TC_ID	MA.EVT.01
Purpose	Fetch supported vision event types and display them for test/configuration.
Requirements	FR-MA-03
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Fetch supported vision event types and display them for test/configuration. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.28 MA.ACT.01

Field	Description
TC_ID	MA.ACT.01
Purpose	Fetch supported backend actions and display available action options.
Requirements	FR-MA-04
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Fetch supported backend actions and display available action options. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.29 MA.MAP.01

Field	Description
TC_ID	MA.MAP.01
Purpose	Retrieve current mappings in the configuration screen.
Requirements	FR-MA-05
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Retrieve current mappings in the configuration screen. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.30 MA.MAP.02

Field	Description
TC_ID	MA.MAP.02
Purpose	Update mappings from the app and verify backend reflects the new mapping.
Requirements	FR-MA-06
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Update mappings from the app and verify backend reflects the new mapping. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.31 MA.TEST.01

Field	Description
TC_ID	MA.TEST.01
Purpose	Send a test vision event from the app and verify backend action execution.
Requirements	FR-MA-07
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Send a test vision event from the app and verify backend action execution. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.32 MA.MAN.01

Field	Description
TC_ID	MA.MAN.01
Purpose	Manually execute LIGHT_ON/LIGHT_OFF actions from the app.
Requirements	FR-MA-08
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Manually execute LIGHT_ON/LIGHT_OFF actions from the app. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.33 MA.ACC.01

Field	Description
TC_ID	MA.ACC.01
Purpose	Verify accessible mode with enlarged controls, high contrast layout, and understandable labels.
Requirements	NFR-MA-01
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify accessible mode with enlarged controls, high contrast layout, and understandable labels. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.34 MA.TTS.01

Field	Description
TC_ID	MA.TTS.01
Purpose	Verify Turkish TTS feedback is spoken after device actions.
Requirements	NFR-MA-02
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify Turkish TTS feedback is spoken after device actions. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.35 MA.ERR.01

Field	Description
TC_ID	MA.ERR.01
Purpose	Verify backend connection failure is shown to the user without app crash.
Requirements	NFR-MA-03
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify backend connection failure is shown to the user without app crash. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.36 SD.MOCK.01

Field	Description
TC_ID	SD.MOCK.01
Purpose	Run backend without SMART_BULB_HOST and verify mock mode works.
Requirements	FR-SD-01
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Run backend without SMART_BULB_HOST and verify mock mode works. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.37 SD.CFG.01

Field	Description
TC_ID	SD.CFG.01
Purpose	Configure SMART_BULB_HOST and credentials and verify controller mode is python-kasa.
Requirements	FR-SD-02
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Configure SMART_BULB_HOST and credentials and verify controller mode is python-kasa. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.38 SD.LGT.01

Field	Description
TC_ID	SD.LGT.01
Purpose	Execute LIGHT_ON and verify backend state and physical bulb on response.
Requirements	FR-SD-03
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Backend configured with SMART_BULB_HOST and credentials; bulb on same LAN.
Procedure	1. Send LIGHT_ON by /actions/execute or mapped event. 2. Observe backend response. 3. Observe physical bulb.
Expected Result / Verification	Backend device_status.light becomes on and the physical bulb turns on.
Cleanup	Return system to default state if needed.

7.39 SD.LGT.02

Field	Description
TC_ID	SD.LGT.02
Purpose	Execute LIGHT_OFF and verify backend state and physical bulb off response.
Requirements	FR-SD-04
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Execute LIGHT_OFF and verify backend state and physical bulb off response. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.40 SD.PLG.01

Field	Description
TC_ID	SD.PLG.01
Purpose	Execute PLUG_ON and verify backend plug state updates.
Requirements	FR-SD-05
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Execute PLUG_ON and verify backend plug state updates. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.41 SD.PLG.02

Field	Description
TC_ID	SD.PLG.02
Purpose	Execute PLUG_OFF and verify backend plug state updates.
Requirements	FR-SD-06
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Execute PLUG_OFF and verify backend plug state updates. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.42 SD.NOB.01

Field	Description
TC_ID	SD.NOB.01
Purpose	Execute device actions when hardware is absent and verify no crash occurs.
Requirements	NFR-SD-01
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Execute device actions when hardware is absent and verify no crash occurs. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.43 EI.EYE.01

Field	Description
TC_ID	EI.EYE.01
Purpose	Eye look_right event navigates UI and updates mobile status.
Requirements	FR-EI-01
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Eye look_right event navigates UI and updates mobile status. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.44 EI.EYE.02

Field	Description
TC_ID	EI.EYE.02
Purpose	Short blink selects Light menu and executes configured action.
Requirements	FR-EI-02
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Short blink selects Light menu and executes configured action. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.45 EI.EYE.03

Field	Description
TC_ID	EI.EYE.03
Purpose	Long blink returns to main menu or cancels current selection.
Requirements	FR-EI-03
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Long blink returns to main menu or cancels current selection. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.46 EI.DEV.01

Field	Description
TC_ID	EI.DEV.01
Purpose	Gesture mapped to LIGHT_ON turns real smart bulb on.
Requirements	FR-EI-04
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Camera, Raspberry Pi, backend, mobile app, and smart bulb are running on the same LAN.
Procedure	1. Map a supported gesture to LIGHT_ON. 2. Perform the gesture. 3. Observe Pi log, backend state, mobile status, and bulb.
Expected Result / Verification	Gesture generates event, backend maps it to LIGHT_ON, mobile status updates, and smart bulb turns on.
Cleanup	Return system to default state if needed.

7.47 EI.DEV.02

Field	Description
TC_ID	EI.DEV.02
Purpose	Gesture mapped to LIGHT_OFF turns real smart bulb off.
Requirements	FR-EI-05
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Gesture mapped to LIGHT_OFF turns real smart bulb off. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.48 EI.MAP.01

Field	Description
TC_ID	EI.MAP.01
Purpose	Update mapping in mobile app and verify next Raspberry Pi event uses new mapping.
Requirements	FR-EI-06
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Update mapping in mobile app and verify next Raspberry Pi event uses new mapping. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.49 EI.HND.01

Field	Description
TC_ID	EI.HND.01
Purpose	Hand gesture event triggers mapped navigation/action in final system.
Requirements	FR-EI-07
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Hand gesture event triggers mapped navigation/action in final system. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.50 EI.AUTH.01

Field	Description
TC_ID	EI.AUTH.01
Purpose	Unauthorized face does not gain device-control access.
Requirements	FR-EI-08
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Unauthorized face does not gain device-control access. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.51 APR.RT.01

Field	Description
TC_ID	APR.RT.01
Purpose	Verify gesture-to-action response feels real-time on Raspberry Pi 5 during demo.
Requirements	NFR-APR-01
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify gesture-to-action response feels real-time on Raspberry Pi 5 during demo. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.52 APR.PR.01

Field	Description
TC_ID	APR.PR.01
Purpose	Verify camera frames are processed locally and not uploaded to cloud services.
Requirements	NFR-APR-02
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Review runtime configuration and network flow. 2. Run vision module. 3. Confirm only event payloads, not video frames, are sent to backend.
Expected Result / Verification	Camera frames are processed locally on Raspberry Pi; no cloud upload is required for operation.
Cleanup	Return system to default state if needed.

7.53 APR.DIS.01

Field	Description
TC_ID	APR.DIS.01
Purpose	Disconnect camera stream and verify graceful error handling.
Requirements	NFR-APR-03
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Disconnect camera stream and verify graceful error handling. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.54 APR.NET.01

Field	Description
TC_ID	APR.NET.01
Purpose	Disconnect backend/network and verify mobile app shows connection
Requirements	NFR-APR-04
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Disconnect backend/network and verify mobile app shows connection error. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.55 APR.FP.01

Field	Description
TC_ID	APR.FP.01
Purpose	Verify stable-gaze duration and blink thresholds reduce false triggers.
Requirements	NFR-APR-05
Priority	High
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify stable-gaze duration and blink thresholds reduce false triggers. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.56 APR.ACC.01

Field	Description
TC_ID	APR.ACC.01
Purpose	Verify hands-free use path is achievable without physical buttons.
Requirements	NFR-APR-06
Priority	Medium
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify hands-free use path is achievable without physical buttons. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.

7.57 APR.LOG.01

Field	Description
TC_ID	APR.LOG.01
Purpose	Verify FPS, gaze state, and event logs are available for debugging.
Requirements	NFR-APR-07
Priority	Low
Estimated Time Needed	5-10 minutes
Dependency	Required module is installed and configured.
Setup	Start related service/component and prepare local network environment.
Procedure	1. Prepare the component under test. 2. Execute scenario: Verify FPS, gaze state, and event logs are available for debugging. 3. Observe response, logs, UI state, and hardware state when applicable.
Expected Result / Verification	Actual result matches the scenario expectation; no crash occurs; returned JSON/UI/device state is consistent.
Cleanup	Return system to default state if needed.